NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Representations of Lie Groups and Lie Algebras: A Mathematical Dissection of Quantum Spin

Chan Lok Hin Gordon (U2440785A)

Abstract

Introductory quantum mechanics courses often present the concept of spin heuristically. In this project, we investigate the mathematical foundations of quantum mechanical spin. In particular, we explore the geometric and algebraic origins of spin, in the language of Lie groups, Lie algebras, and their unitary and irreducible representations. A central focus is the topological and algebraic relationship between the rotation group $\mathsf{SO}(3)$ and its universal double cover, the special unitary group $\mathsf{SU}(2)$, which is diffeomorphic to $\mathbb{S}^3$. By classifying the irreducible representations of $\mathsf{SU}(2)$, we present the formalism of how this abstract algebraic classification naturally gives rise to the physical phenomenon of integer and half-integer spin. Ultimately, we hope to bridge topology and abstract algebra with the physical ideas in quantum physics.

Topological Spaces $(X, \mathcal{T})$

set X, $\quad \mathcal{T} \subseteq \mathscr{P}(X)$:

1. (Trivial Subsets) $\varnothing \in \mathcal{T}$ and $X \in \mathcal{T}$.
2. (Arbitrary Unions) $\forall \{U_\alpha\}_{\alpha\in\mathcal{I}} \subseteq \mathcal{T} : \bigcup_{\alpha\in\mathcal{I}} U_\alpha \in \mathcal{T}$.
3. (Finite Intersections) $\forall U, V \in \mathcal{T} : U \cap V \in \mathcal{T}$.

Topological Manifolds $(M, \mathcal{T})$

topological space $(M, \mathcal{T})$:

1. (Hausdorffness) $\forall p \in M, \forall q \in M \setminus \{p\}, \exists U, V \in \mathcal{T}$: $p \in U$ and $q \in V$ and $U \cap V = \varnothing$.
2. (Second-Countability) $\exists \{B_i\}_{i\in\mathbb{N}} \subseteq \mathcal{T}, \forall U \in \mathcal{T}, \forall x \in U, \exists i \in \mathbb{N}$: $x \in B_i \subseteq U$.
3. (Local Euclideanness) $\forall p \in M, \exists U \in \{U \in \mathcal{T} \mid p \in U\}, \exists V \in \mathcal{T}_{\mathbb{R}^n}$: $(U, \mathcal{T}|_U) \cong (V, \mathcal{T}_{\mathbb{R}^n}|_V)$.

Smooth Manifolds $(M, \mathcal{T}, \mathcal{A})$

topological manifold $(M, \mathcal{T})$, $\quad \mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha\in\mathcal{I}}$:

1. (Atlas) $\forall \alpha \in \mathcal{I}$: $(U_\alpha, \varphi_\alpha)$ is an $(M, \mathcal{T})$-chart and $\bigcup_{\alpha\in\mathcal{I}} U_\alpha = M$.
2. (Smoothness) $\forall \alpha, \beta \in \mathcal{I}$: $\quad \varphi_\beta \circ \varphi_\alpha^{-1}$ is C^∞.
3. (Maximality) $\mathcal{A}$ is maximal w.r.t. smoothness.

Tangent Spaces T_pM

smooth manifold M, $p \in M$, $C_p^\infty M = \mathrm{Rep}_p M / \sim_p$:

$$T_pM = \{v : C_p^\infty M \to \mathbb{R} \mid v \text{ is a derivation at } p\}.$$

Groups $(G, \{\mu, \iota, e\})$

set G, $\quad \mu : G \times G \to G$, $\quad \iota : G \to G$, $\quad e \in G$:

1. (Associativity) $\mu(\mu(x, y), z) = \mu(x, \mu(y, z))$.
2. (Identity) $\mu(e, x) = x = \mu(x, e)$.
3. (Inverse) $\mu(\iota(x), x) = e = \mu(x, \iota(x))$.

Lie Groups $(G, \mathcal{T}, \mathcal{A}, \{\mu, \iota, e\})$

smooth manifold $(G, \mathcal{T}, \mathcal{A})$, $\quad$ group $(G, \{\mu, \iota, e\})$:

1. $\mu : G \times G \to G$ is smooth w.r.t. $\mathcal{A} \otimes \mathcal{A}$ and $\mathcal{A}$.
2. $\iota : G \to G$ is smooth w.r.t. $\mathcal{A}$ and $\mathcal{A}$.

Lie $\mathbb{F}$-Algebra $(\mathfrak{g}, \{+, -, \mathbf{0}, \cdot, [\bullet, \bullet]\})$

$\mathbb{F}$-vector space $(\mathfrak{g}, \{+, -, \mathbf{0}, \cdot\})$, $\quad [\bullet, \bullet] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$:

1. (Bilinearity) $\forall X, Y, Z \in \mathfrak{g}, \forall \lambda \in \mathbb{F}$:
$$[X + \lambda \cdot Y, Z] = [X, Z] + \lambda \cdot [Y, Z];$$
$$[X, Y + \lambda \cdot Z] = [X, Y] + \lambda \cdot [X, Z].$$
2. (Skew-symmetry) $\forall X, Y \in \mathfrak{g} : [Y, X] = -[X, Y]$.
3. (Jacobi Identity) $\forall X, Y, Z \in \mathfrak{g}$:
$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = \mathbf{0}.$$

Tangent Spaces at Identity are Lie Algebras

Let $\mathfrak{g} = T_eG$ be the tangent space of a Lie group G at the identity. For each $v \in \mathfrak{g}$, pick the unique left-invariant vector field $X^v \in \mathfrak{X}_L G$ with $X^v|_e = v$. For any $v, w \in \mathfrak{g}$, define $[v, w]_\mathfrak{g} = [X^v, X^w]_\circ|_e$. This makes $(\mathfrak{g}, [\bullet, \bullet]_\mathfrak{g})$ the Lie algebra of G. Moreover, we have $\dim(\mathfrak{g}) = \dim(G)$.

Closed Subgroup Theorem – É. Cartan

Given a Lie group $(G, \mathcal{T}, \mathcal{A}, \{\mu, \iota, e\})$ and a closed subgroup $H \subseteq G$, there exists a unique smooth structure $\mathcal{A}_H$ satisfying:

1. $\iota_H : H \hookrightarrow G$ is a smooth embedding.
2. $(H, \mathcal{T}|_H, \mathcal{A}_H, \mu|_{H\times H}, \iota|_H, e)$ is a Lie group.

The General Linear Group

$\mathsf{GL}(n; \mathbb{F}) = \{\boldsymbol{A} \in \mathbb{F}^{n,n} \mid \det(\boldsymbol{A}) \neq 0\} = \det^{-1}(\mathbb{F} \setminus \{0\})$

Since $\det : \mathbb{F}^{n,n} \to \mathbb{F}$ is continuous, $\mathsf{GL}(n; \mathbb{F})$ is open in $\mathbb{F}^{n,n}$, making it a Lie group with the substructure.

The Special Orthogonal Group (Rigid Rotations)

$$\mathsf{SO}(3) = \{\boldsymbol{R} \in \mathbb{R}^{3,3} \mid \boldsymbol{R}^\intercal \boldsymbol{R} = \boldsymbol{I}_3 \text{ and } \det(\boldsymbol{R}) = 1\};$$
$$\mathfrak{so}(3) = T_{\boldsymbol{I}_3}\mathsf{SO}(3) = \{\boldsymbol{R} \in \mathbb{R}^{3,3} \mid \boldsymbol{R}^\intercal = -\boldsymbol{R}\}.$$

$\mathsf{SO}(3)$ is a closed Lie subgroup of $\mathsf{GL}(3; \mathbb{R})$.

$\dim(\mathfrak{so}(3)) = \dim(\mathsf{SO}(3)) = 3$, with basis $[\boldsymbol{L}_i, \boldsymbol{L}_j] = \varepsilon_{ijk}\boldsymbol{L}_k$:

$$(\boldsymbol{L}_1, \boldsymbol{L}_2, \boldsymbol{L}_3) = \left(\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\right).$$

The Special Unitary Group

$$\mathsf{SU}(2) = \{\boldsymbol{U} \in \mathbb{C}^{2,2} \mid \boldsymbol{U}^\dagger \boldsymbol{U} = \boldsymbol{I}_2 \text{ and } \det(\boldsymbol{U}) = 1\};$$
$$\mathfrak{su}(2) = T_{\boldsymbol{I}_2}\mathsf{SU}(2) = \{\boldsymbol{U} \in \mathbb{C}^{2,2} \mid \boldsymbol{U}^\dagger = -\boldsymbol{U}, \mathrm{tr}(\boldsymbol{U}) = 0\}.$$

$\mathsf{SU}(2)$ is a closed Lie subgroup of $\mathsf{GL}(2; \mathbb{C})$.

$\dim(\mathfrak{su}(2)) = \dim(\mathsf{SU}(2)) = 3$, with basis $[\boldsymbol{T}_i, \boldsymbol{T}_j] = \varepsilon_{ijk}\boldsymbol{T}_k$:

$$(\boldsymbol{T}_1, \boldsymbol{T}_2, \boldsymbol{T}_3) = \left(\frac{1}{2}\begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}, \frac{1}{2}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \frac{1}{2}\begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}\right).$$

The Double Cover $\Phi : \mathsf{SU}(2) \to \mathsf{SO}(3)$

For each $\boldsymbol{U} \in \mathsf{SU}(2)$, define a linear map $\Phi(\boldsymbol{U}) : \mathfrak{su}(2) \to \mathfrak{su}(2)$ by $[\Phi(\boldsymbol{U})](\boldsymbol{X}) = \boldsymbol{U}\boldsymbol{X}\boldsymbol{U}^\dagger$. Coordinatize $\mathfrak{su}(2)$ by $\beta = (\boldsymbol{T}_1, \boldsymbol{T}_2, \boldsymbol{T}_3)$. One verifies:

- $[\Phi(\boldsymbol{U})]_\beta^\beta \in \mathsf{SO}(3)$,
- $\Phi(\boldsymbol{U}\boldsymbol{V}) = \Phi(\boldsymbol{U}) \circ \Phi(\boldsymbol{V})$ (smooth homomorphism),
- $\ker(\Phi) = \{\pm \boldsymbol{I}_2\}$, and Φ is surjective.

Thus $\mathsf{SU}(2)$ is a double cover of $\mathsf{SO}(3)$: each rotation in $\mathsf{SO}(3)$ corresponds to exactly two matrices in $\mathsf{SU}(2)$ that differ only by a sign.

Lie Algebra Isomorphism $\mathfrak{so}(3) \cong \mathfrak{su}(2)$

Define $\varphi : \mathfrak{so}(3) \to \mathfrak{su}(2)$ by $\varphi(\boldsymbol{L}_i) = \boldsymbol{T}_i$. φ is a linear isomorphism. Moreover, $\varphi([\boldsymbol{L}_i, \boldsymbol{L}_j]) = \varphi(\varepsilon_{ijk}\boldsymbol{L}_k) = \varepsilon_{ijk}\varphi(\boldsymbol{L}_k) = \varepsilon_{ijk}\boldsymbol{T}_k = [\boldsymbol{T}_i, \boldsymbol{T}_j] = [\varphi(\boldsymbol{L}_i), \varphi(\boldsymbol{L}_j)]$.

Lie Algebra Representations $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$

Given a Lie $\mathbb{F}$-algebra $\mathfrak{g}$ and an $\mathbb{F}$-vector space V, a representation of $\mathfrak{g}$ on V is a Lie algebra homomorphism $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ that preserves the bracket:

$$\forall X, Y \in \mathfrak{g} : \quad \pi\left([X, Y]_\mathfrak{g}\right) = [\pi(X), \pi(Y)]_\circ.$$

The representation is irreducible iff the only subspaces $W \subseteq V$ with $[\pi(X)](W) \subseteq W$ for *all* $X \in \mathfrak{g}$ are the trivial ones: $\{\mathbf{0}\}$ and V itself.

Lie Group Representations $\Pi : G \to \mathsf{GL}(V)$

Given a Lie group G and an $\mathbb{F}$-vector space V, an ordinary representation of G on V is a smooth group homomorphism $\Pi : G \to \mathsf{GL}(V)$. The representation is irreducible iff the only subspaces $W \subseteq V$ with $[\Pi(g)](W) \subseteq W$ for all $g \in G$ are the trivial ones; it is unitary iff $\Pi(g)^\dagger \Pi(g) = I_V$ for all $g \in G$.

A projective representation is a smooth map $\rho : G \to \mathsf{GL}(V)$ with a smooth function $\omega : G \times G \to \mathbb{F}^\times$ such that $\rho(g)\rho(h) = \omega(g, h)\rho(gh)$ and $\rho(e) = I_V$.

Complexification of a Real Lie Algebra

Let $\mathfrak{g}$ be a Lie $\mathbb{R}$-algebra. Its complexification, denoted by "$\mathfrak{g}_\mathbb{C}$", is defined as $\mathfrak{g} \otimes_\mathbb{R} \mathbb{C} = \{X + iY \mid X, Y \in \mathfrak{g}\}$, which is a Lie $\mathbb{C}$-algebra with the bracket extended $\mathbb{C}$-bilinearly.

Complex Lie Algebra Representation

Let $\mathfrak{g}$ be a Lie $\mathbb{R}$-algebra. A complex representation of $\mathfrak{g}$ is a representation $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ where V is a $\mathbb{C}$-vector space. Equivalently, π extends uniquely to a representation of the complexified algebra $\mathfrak{g}_\mathbb{C}$.

Complexification and Ladder Operators

$$\mathfrak{su}(2)_\mathbb{C} = \mathfrak{sl}(2, \mathbb{C}) = \{\boldsymbol{A} \in \mathbb{C}^{2,2} \mid \mathrm{tr}(\boldsymbol{A}) = 0\}$$

Cartan-Weyl basis:

$$(\boldsymbol{H}, \boldsymbol{E}, \boldsymbol{F}) = (2i\boldsymbol{T}_3, i\boldsymbol{T}_1 - \boldsymbol{T}_2, i\boldsymbol{T}_1 + \boldsymbol{T}_2) = \left(\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right).$$

$$[\boldsymbol{H}, \boldsymbol{E}] = 2\boldsymbol{E}, \qquad [\boldsymbol{H}, \boldsymbol{F}] = -2\boldsymbol{F}, \qquad [\boldsymbol{E}, \boldsymbol{F}] = \boldsymbol{H}.$$

Irreducible Representations of $\mathfrak{su}(2)$

Every finite-dimensional irreducible complex representation of $\mathfrak{su}(2)$ is equivalent to exactly one of the following: for each $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, there exists an irreducible representation $\pi_j : \mathfrak{su}(2) \to \mathfrak{gl}(V_j)$ with $\dim(V_j) = 2j + 1$ and a weight basis $(\boldsymbol{v}_j, \boldsymbol{v}_{j-1}, \dots, \boldsymbol{v}_{-j})$ of V_j such that:

$$[\pi_j(\boldsymbol{H})](\boldsymbol{v}_m) = 2m\boldsymbol{v}_m,$$
$$[\pi_j(\boldsymbol{E})](\boldsymbol{v}_m) = \sqrt{(j-m)(j+m+1)}\ \boldsymbol{v}_{m+1},$$
$$[\pi_j(\boldsymbol{F})](\boldsymbol{v}_m) = \sqrt{(j+m)(j-m+1)}\ \boldsymbol{v}_{m-1},$$

with the convention $\boldsymbol{v}_{-j-1} = \boldsymbol{v}_{j+1} = \mathbf{0}_{V_j}$.

j	$\dim(V_j)$	Name
0	1	scalar
$\frac{1}{2}$	2	spin-$\frac{1}{2}$ (spinor)
1	3	vector (spin-1)
$\frac{3}{2}$	4	spin-$\frac{3}{2}$
$\vdots$	$\vdots$	$\vdots$

Integration to the Group $\mathsf{SU}(2)$

$\mathsf{SU}(2) \cong \mathbb{S}^3$ is simply connected. Hence every Lie algebra representation $\pi_j : \mathfrak{su}(2) \to \mathfrak{gl}(V_j)$ integrates uniquely to a Lie group representation

$$\Pi_j : \mathsf{SU}(2) \longrightarrow \mathsf{GL}(V_j), \qquad \Pi_j(e^{\boldsymbol{X}}) = e^{\pi_j(\boldsymbol{X})}.$$

Why $\mathsf{SU}(2)$?

Physical states are rays, so symmetries may be represented only up to a phase. Every projective representation of $\mathsf{SO}(3)$ lifts to an ordinary representation of the double cover $\mathsf{SU}(2)$. Conversely, an irreducible ordinary representation Π_j of $\mathsf{SU}(2)$ descends to an ordinary representation of $\mathsf{SO}(3)$ iff $\Pi_j(-\boldsymbol{I}_2) = I_{V_j}$; this holds exactly for integer j.

Descent to $\mathsf{SO}(3)$: Integer vs. Half-Integer Spin

For the double cover $\Phi : \mathsf{SU}(2) \to \mathsf{SO}(3)$ we have

$$\Pi_j(-\boldsymbol{I}_2) = e^{2\pi \cdot \pi_j(\boldsymbol{T}_3)} = (-1)^{2j} I_{V_j}.$$

Consequently:

- **(Integer j)** $\Pi_j(-\boldsymbol{I}_2) = I \implies \Pi_j$ factors through Φ, giving an ordinary representation of $\mathsf{SO}(3)$ (integer spin).
- **(Half-integer j)** $\Pi_j(-\boldsymbol{I}_2) = -I \implies \Pi_j$ does not descend; it yields a projective representation of $\mathsf{SO}(3)$ (spinor representation).

Physical Spin Operators

Define the Hermitian spin observables $S_k = i\hbar\pi_j(\boldsymbol{T}_k)$. They satisfy the angular-momentum algebra $[S_i, S_j] = i\hbar\varepsilon_{ijk}S_k$. S_z has eigenvalues $\hbar m$ $(m = j, \dots, -j)$. For $j = \frac{1}{2}$, with $\beta = \left(\boldsymbol{v}_{\frac{1}{2}}, \boldsymbol{v}_{-\frac{1}{2}}\right)$, $[S_k]_\beta^\beta = \frac{\hbar}{2}\boldsymbol{\sigma}_k$ (Pauli matrices).

The Spinor Sign Change (4π Periodicity)

A rotation by 2π lifts to $-\boldsymbol{I}_2 \in \mathsf{SU}(2)$. In a half-integer spin representation $\Pi_j(-\boldsymbol{I}_2) = -I$, so the state vector flips sign. The original sign is restored only after a 4π rotation. This is the hallmark of spinors in quantum mechanics.

Foundations

All definitions and theorems are proven rigorously in the accompanying project document. Scan the QR code or click below to read the full paper with proofs.

Full paper with proofs (PDF)

Supervisor: **Assoc. Prof. François Gay-Balmaz**